

# Newton-Raphson Method

Taylor Series for Function of Several Variables around A Point  $(a_1, a_2, \dots, a_n)$ :

$$f_i(x_1, x_2, \dots, x_n) = \sum_{k_1=0}^{\infty} \dots \sum_{k_n=0}^{\infty} \frac{(x_1 - a_1)^{k_1} \dots (x_n - a_n)^{k_n}}{(k_1!) \dots (k_n!)} \cdot \left[ \frac{\partial^{k_1+\dots+k_n}}{\partial x_1^{k_1} \dots \partial x_n^{k_n}} f_i(a_1, a_2, \dots, a_n) \right]$$

Taylor Series Approximation:

$$\begin{aligned} f_i(x_1, x_2, \dots, x_n) \cong & f_i(a_1, a_2, \dots, a_n) + \left[ \frac{\partial}{\partial x_1} f_i(a_1, a_2, \dots, a_n) \right] \cdot (x_1 - a_1) \\ & + \left[ \frac{\partial}{\partial x_2} f_i(a_1, a_2, \dots, a_n) \right] \cdot (x_2 - a_2) + \dots + \left[ \frac{\partial}{\partial x_n} f_i(a_1, a_2, \dots, a_n) \right] \cdot (x_n - a_n) \end{aligned}$$

Rearranging:

$$\begin{aligned} f_i(x_1, x_2, \dots, x_n) = 0 \quad \rightarrow \quad & \left[ \frac{\partial}{\partial x_1} f_i(a_1, a_2, \dots, a_n) \right] \cdot (x_1 - a_1) + \left[ \frac{\partial}{\partial x_2} f_i(a_1, a_2, \dots, a_n) \right] \cdot (x_2 - a_2) \\ & + \dots + \left[ \frac{\partial}{\partial x_n} f_i(a_1, a_2, \dots, a_n) \right] \cdot (x_n - a_n) = -f_i(a_1, a_2, \dots, a_n) \end{aligned}$$

# Newton-Raphson Method

Matrix Form:

$$\begin{array}{l} x_1 - a_1 = \Delta x_1 \\ x_2 - a_2 = \Delta x_2 \\ \vdots \\ x_n - a_n = \Delta x_n \end{array} \rightarrow \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \begin{Bmatrix} \Delta x_1 \\ \Delta x_2 \\ \vdots \\ \Delta x_n \end{Bmatrix} = - \begin{Bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{Bmatrix} \rightarrow \mathbf{J}_{\partial \mathbf{f} / \partial \mathbf{x}} \cdot \Delta \mathbf{x} = -\mathbf{f}(\mathbf{x})$$

$\mathbf{J}_{\partial \mathbf{f} / \partial \mathbf{x}}$  is Jacobian Matrix

Recurrence Relation:

$$\mathbf{J}_{\partial \mathbf{f} / \partial \mathbf{x}}^{(k-1)} \cdot \mathbf{d}^{(k)} = -\mathbf{f}(\mathbf{x}^{(k-1)})$$
$$\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} + \mathbf{d}^{(k)}$$

$\mathbf{d} = \Delta \mathbf{x}$  is search direction.

Relative Error:

$$\varepsilon^{(k)} = \frac{\|\mathbf{x}^{(k-1)} - \mathbf{x}^{(k)}\|_2}{\|\mathbf{x}^{(k)}\|_2} = \frac{\|\mathbf{d}^{(k)}\|_2}{\|\mathbf{x}^{(k)}\|_2}$$

# Newton-Raphson Method

Example 03-08: Find root of the following system of equations using Newton-Raphson method with initial guess  $\mathbf{x}^{(0)} = \mathbf{1}$  and  $\varepsilon_t = 10^{-6}$  for relative error.

$$x_1 + 2x_2 + x_3 + 4x_4 = 20.700$$

$$x_1^2 + 2x_1x_2 + x_4^3 = 15.880$$

$$x_1^3 + x_3^2 + x_4 = 21.218$$

$$3x_2 + x_3x_4 = 21.100$$

→

$$f_1(x_1, x_2, x_3, x_4) = x_1 + 2x_2 + x_3 + 4x_4 - 20.700 = 0$$

$$f_2(x_1, x_2, x_3, x_4) = x_1^2 + 2x_1x_2 + x_4^3 - 15.880 = 0$$

$$f_3(x_1, x_2, x_3, x_4) = x_1^3 + x_3^2 + x_4 - 21.218 = 0$$

$$f_4(x_1, x_2, x_3, x_4) = 3x_2 + x_3x_4 - 21.100 = 0$$

$$\mathbf{J}_{\partial f / \partial \mathbf{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} & \frac{\partial f_1}{\partial x_4} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} & \frac{\partial f_2}{\partial x_4} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} & \frac{\partial f_3}{\partial x_4} \\ \frac{\partial f_4}{\partial x_1} & \frac{\partial f_4}{\partial x_2} & \frac{\partial f_4}{\partial x_3} & \frac{\partial f_4}{\partial x_4} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 & 4 \\ 2x_1 + 2x_2 & 2x_1 & 0 & 3x_4^2 \\ 3x_1^2 & 0 & 2x_3 & 1 \\ 0 & 3 & x_4 & x_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 & 4 \\ 2x_1 + 2x_2 & 2x_1 & 0 & 3x_4^2 \\ 3x_1^2 & 0 & 2x_3 & 1 \\ 0 & 3 & x_4 & x_3 \end{bmatrix} \begin{Bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta x_3 \\ \Delta x_4 \end{Bmatrix} = - \begin{Bmatrix} x_1 + 2x_2 + x_3 + 4x_4 - 20.700 \\ x_1^2 + 2x_1x_2 + x_4^3 - 15.880 \\ x_1^3 + x_3^2 + x_4 - 21.218 \\ 3x_2 + x_3x_4 - 21.100 \end{Bmatrix}$$

# Newton-Raphson Method

Example 03-08: Find root of the following system of equations using Newton-Raphson method with initial guess  $\mathbf{x}^{(0)} = \mathbf{1}$  and  $\varepsilon_t = 10^{-6}$  for relative error.

$$x_1 + 2x_2 + x_3 + 4x_4 = 20.700$$

$$x_1^2 + 2x_1x_2 + x_4^3 = 15.880$$

$$x_1^3 + x_3^2 + x_4 = 21.218$$

$$3x_2 + x_3x_4 = 21.100$$

→

$$f_1(x_1, x_2, x_3, x_4) = x_1 + 2x_2 + x_3 + 4x_4 - 20.700 = 0$$

$$f_2(x_1, x_2, x_3, x_4) = x_1^2 + 2x_1x_2 + x_4^3 - 15.880 = 0$$

$$f_3(x_1, x_2, x_3, x_4) = x_1^3 + x_3^2 + x_4 - 21.218 = 0$$

$$f_4(x_1, x_2, x_3, x_4) = 3x_2 + x_3x_4 - 21.100 = 0$$

$$\mathbf{J}_{\partial f / \partial \mathbf{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} & \frac{\partial f_1}{\partial x_4} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} & \frac{\partial f_2}{\partial x_4} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} & \frac{\partial f_3}{\partial x_4} \\ \frac{\partial f_4}{\partial x_1} & \frac{\partial f_4}{\partial x_2} & \frac{\partial f_4}{\partial x_3} & \frac{\partial f_4}{\partial x_4} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 & 4 \\ 2x_1 + 2x_2 & 2x_1 & 0 & 3x_4^2 \\ 3x_1^2 & 0 & 2x_3 & 1 \\ 0 & 3 & x_4 & x_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 & 4 \\ 2x_1 + 2x_2 & 2x_1 & 0 & 3x_4^2 \\ 3x_1^2 & 0 & 2x_3 & 1 \\ 0 & 3 & x_4 & x_3 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{bmatrix} = - \begin{bmatrix} x_1 + 2x_2 + x_3 + 4x_4 - 20.700 \\ x_1^2 + 2x_1x_2 + x_4^3 - 15.880 \\ x_1^3 + x_3^2 + x_4 - 21.218 \\ 3x_2 + x_3x_4 - 21.100 \end{bmatrix}$$

# Newton-Raphson Method

Example 03-08: Find root of the following system of equations using Newton-Raphson method with initial guess  $\mathbf{x}^{(0)} = \mathbf{1}$  and  $\varepsilon_t = 10^{-6}$  for relative error.

$$x_1 + 2x_2 + x_3 + 4x_4 = 20.700$$

$$x_1^2 + 2x_1x_2 + x_4^3 = 15.880$$

$$x_1^3 + x_3^2 + x_4 = 21.218$$

$$3x_2 + x_3x_4 = 21.100$$

→

$$f_1(x_1, x_2, x_3, x_4) = x_1 + 2x_2 + x_3 + 4x_4 - 20.700 = 0$$

$$f_2(x_1, x_2, x_3, x_4) = x_1^2 + 2x_1x_2 + x_4^3 - 15.880 = 0$$

$$f_3(x_1, x_2, x_3, x_4) = x_1^3 + x_3^2 + x_4 - 21.218 = 0$$

$$f_4(x_1, x_2, x_3, x_4) = 3x_2 + x_3x_4 - 21.100 = 0$$

$$\mathbf{J}_{\partial \mathbf{f} / \partial \mathbf{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} & \frac{\partial f_1}{\partial x_4} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} & \frac{\partial f_2}{\partial x_4} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} & \frac{\partial f_3}{\partial x_4} \\ \frac{\partial f_4}{\partial x_1} & \frac{\partial f_4}{\partial x_2} & \frac{\partial f_4}{\partial x_3} & \frac{\partial f_4}{\partial x_4} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 & 4 \\ 2x_1 + 2x_2 & 2x_1 & 0 & 3x_4^2 \\ 3x_1^2 & 0 & 2x_3 & 1 \\ 0 & 3 & x_4 & x_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 & 4 \\ 2x_1 + 2x_2 & 2x_1 & 0 & 3x_4^2 \\ 3x_1^2 & 0 & 2x_3 & 1 \\ 0 & 3 & x_4 & x_3 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{bmatrix} = - \begin{bmatrix} x_1 + 2x_2 + x_3 + 4x_4 - 20.700 \\ x_1^2 + 2x_1x_2 + x_4^3 - 15.880 \\ x_1^3 + x_3^2 + x_4 - 21.218 \\ 3x_2 + x_3x_4 - 21.100 \end{bmatrix} \rightarrow \begin{aligned} \mathbf{x}^{(k)} &= \mathbf{x}^{(k-1)} + \mathbf{d}^{(k)} \\ \mathbf{J}_{\partial \mathbf{f} / \partial \mathbf{x}}^{(k)} \cdot \mathbf{d}^{(k+1)} &= -\mathbf{f}(\mathbf{x}^{(k)}) \\ &\vdots \end{aligned}$$